SEGAN

Complete Installation, Execution & Scientific Reproduction Manual

Windows-first | Numbered, verifiable instructions | MATLAB + MATPOWER + Gurobi | Quick and Full Mode

PURPOSE

A new researcher should be able to download the complete SEGAN repository, install prerequisites, activate the optimizer, regenerate the supported article figures/tables, and understand which numerical checks establish reproducibility.

Associated article

Feyyaz Fatih Aydin (2026), "Auditable state-based screening of capacity-withholding effects in nodal electricity markets," Sustainable Energy, Grids and Networks, article 102558. DOI: 10.1016/j.segan.2026.102558.

Repository: github.com/feyyazfatihaydin/SEGAN-State-Based-Screening

Document status: Release-ready documentation edition, 04 November 2026. Based on the provided SEGAN README, recorded author validations and cited official installation/licensing sources. Illustrated checkpoints are schematic, not actual third-party installer captures; independent end-user execution was not performed while preparing this manual.

IMPORTANT COMPATIBILITY FINDING

The study README identifies MATLAB R2025b Update 3 and Gurobi 13.0.1 in the reported environment. Gurobi's published support matrix lists MATLAB only through R2025a for Gurobi 13.0.1, while 13.0.2/13.0.3 list R2025b. The manual preserves the study record and separately recommends an officially supported combination for new installations. Solver/version changes may change numerical details.

How to use this manual

Complete Stages 01-08 first. Then choose Track Q for Quick Mode or Track F for Full Mode. Every stage ends with an objective success condition. If a condition fails, use the corresponding troubleshooting table and do not proceed blindly.

Part	Stages	Outcome
Prepare	01-03	Download, inventory and MATLAB installation
Connect tools	04-08	MATPOWER, Gurobi license and MATLAB solver checks
Track Q	09-10	Use existing 12 MAT checkpoints to export 27 files
Track F	11-13	Create isolated workspace; run 23 computation/export stages; validate
Scientific audit	14-16	Identify metrics, figures, limits and reproducibility evidence
Troubleshooting	17	Resolve common failures and produce evidence checklist

WHERE TO TYPE COMMANDS

Windows CMD: prompt normally starts with C:\...> and is used for filesystem/optional Gurobi licensing commands. MATLAB Command Window: prompt starts with >> and is used for install_matpower, gurobi_setup and every SEGAN .m command. Do not paste MATLAB code into CMD.

Expected research outputs

Item	Scope
IEEE 5-bus	16 scenarios x 24 hours = 384 hourly records
IEEE 57-bus	Single-G5 and Double-G5/G7: 16 x 2,160 hours each = 34,560 + 34,560
Combined main library	69,504 hourly records; separate generator-selection screening not in this count
Publication exports	Figures 2-7 and Tables 4-9; 27 files when counting formats and CSV exports
Not regenerated	Figure 1 (manually drawn); Tables 1-2 descriptive; Table 3 settings

STAGE 01 / Download the repository

Goal: Obtain all bundled source files and preserve their relative locations.

Open: Browser + Windows File Explorer

1. Open the SEGAN GitHub repository and sign in if it is still private. The owner must make it public before unauthorised readers can download it.

SEGAN repository

2. Select Code → Download ZIP; extract the complete ZIP. For examples below, assume the extracted root is C:\Research\SEGAN-State-Based-Screening (use your actual folder name).

3. Do not copy individual MATLAB files outside their folders. MATLAB package functions must remain in src/+segan/.

SCREEN CHECKPOINT (illustrative; not an actual application capture)

WHAT YOU SHOULD SEE
Repository root: README.md / USER_GUIDE.md / reproduce_all.m
data/raw: 4 canonical inputs
outputs/cache: 12 .mat checkpoints
src/+segan: modular SEGAN MATLAB functions

> **PASS**
>
> All four inputs and twelve checkpoints exist. No MATLAB/Gurobi binary or licence file is expected in the repository.

STAGE 02 / Verify the input and result inventory

Goal: Make sure the downloaded scientific assets have not been omitted.

Open: Windows CMD

1. Open File Explorer inside the SEGAN root; click the address bar, type cmd and press Enter.

TYPE IN: Windows CMD

```
dir /b data\raw
dir /b outputs\cache\*.mat
dir /b RELEASE_DATA_SHA256.tsv
```

2. Expect these exact raw filenames (capitalisation and spaces matter when typing paths):

[ ] `AMES_5Bus.xlsx`

[ ] `case57.m`

[ ] `pglib_opf_case57_ieee__sad.m`

[ ] `power Generation and consumption.csv`

3. For a basic SHA-256 check, use the published RELEASE_DATA_SHA256.tsv. A file hash can be viewed with:

TYPE IN: Windows CMD - optional example

```
powershell -NoProfile -Command "(Get-FileHash 'data/raw/AMES_5Bus.xlsx' -Algorithm SHA256).Hash"
```

LINE-ENDING NOTE

The two canonical .m input files can suffer Windows LF/CRLF conversion in some clone configurations. The repository includes .gitattributes to preserve them. A bytewise hash mismatch caused exclusively by line endings is not proof of altered numeric coefficients; do not silently edit scientific data to force a match.

STAGE 03 / Install MATLAB and toolbox

Goal: Provide the MATLAB environment needed for the numerical pipeline.

Open: MathWorks installer; MATLAB Command Window

---

1. Sign in to MathWorks and install a properly licensed 64-bit MATLAB edition. The reported study setup is R2025b Update 3; a different release must be disclosed as a reproduction environment difference.

MathWorks downloads

2. During installation, include the Statistics and Machine Learning Toolbox (K-means, PCA and related analyses).

3. Launch MATLAB; enter:

TYPE IN: MATLAB Command Window

```
version
ver
license('test','Statistics_Toolbox')
```

PASS

MATLAB runs and ver shows Statistics and Machine Learning Toolbox; license test returns true for an available licence. Quick Mode may not use the expensive fitting stages, but Full Mode requires the appropriate toolbox.

STAGE 04 / Install MATPOWER 8.1

Goal: Make DC optimal power flow routines accessible to MATLAB.

Open: Web browser + MATLAB Command Window

---

1. Download MATPOWER 8.1 (not an unspecified latest release) from the official download page and extract outside the SEGAN folder.

Official MATPOWER download

2. In MATLAB, change Current Folder to the extracted MATPOWER root (where install_matpower.m is located). Example:

TYPE IN: MATLAB Command Window

```
cd('C:/Tools/matpower8.1')
install_matpower
```

3. Follow the installer's path prompts. Save the MATLAB path if you wish to retain it across sessions. Verify:

TYPE IN: MATLAB Command Window

```
which rundcopf
test_matpower
```

SCREEN CHECKPOINT (illustrative; not an actual application capture)

WHAT YOU SHOULD SEE
MATLAB resolves rundcopf to a MATPOWER path
MATPOWER test output completes without unexpected failures

> **PASS**
>
> MATPOWER is on the MATLAB search path. If which rundcopf is empty, rerun install_matpower from the correct extracted directory.

STAGE 05 / Install a compatible Gurobi distribution

Goal: Install a Gurobi version with a compatible MATLAB interface.

Open: Browser + Windows installer

---

1. Create/sign in to a Gurobi account. Download the full Windows 64-bit Optimizer package - not a Python-only package.

Official Gurobi installation

2. Choose version carefully. Historical study record: Gurobi 13.0.1. New R2025b users: an officially supported Gurobi 13.0.2 or 13.0.3 combination is preferable; record the choice in your reproduction notes.

Version-specific MATLAB compatibility matrix

3. Run the full MSI installer. For Gurobi 13.0.3, a typical installation root is C:\gurobi1303\win64; actual installation location can differ. Look for a matlab subfolder.

TYPE IN: Windows CMD - example for 13.0.3

```
dir C:\gurobi1303\win64\matlab\gurobi_setup.m
```

PASS

The installation contains matlab/gurobi_setup.m. Installer completion alone does not activate a Gurobi license.

STAGE 06 / Request the academic Gurobi license

Goal: Activate the optimizer under the user's own academic eligibility.

Open: Gurobi portal + Windows CMD

1. Register/sign in as an academic user, use your university affiliation and follow the provider's eligibility instructions. For initial activation, use a university network or suitable institution VPN if required.

Free academic license and eligibility

2. Select one appropriate route: Academic Named-User (one user/machine; offline execution after activation) or Academic WLS (portable; needs an ongoing Internet connection during solves).

Named-User	Academic WLS
Install to an individual machine	Credentials associated with the licensed user
University network required for activation	University network required for initial issue/renewal
Offline solver runs after activation	Online Gurobi license service needed for runs
Use portal-provided grbgetkey	Follow portal WLS configuration; no grbgetkey

3. Named-User: copy the exact grbgetkey command generated by your own user portal. Enter it in Windows CMD, not MATLAB. The example below is a placeholder, not a working key.

TYPE IN: Windows CMD - Named-User only

```
grbgetkey YOUR-OWN-PORTAL-KEY
```

4. Accept the standard location for gurobi.lic when practical. For WLS, follow official WLS-specific instructions instead.

Official named-user activation

CREDENTIAL SAFETY

Never upload grbgetkey tokens, gurobi.lic, WLS API keys or licence screenshots revealing secrets to GitHub, Zenodo or a research support request.

STAGE 07 / Connect Gurobi to MATLAB

Goal: Install the Gurobi MATLAB interface into the MATLAB path and verify that it is recognised.

Open: MATLAB Command Window

---

1. Start MATLAB. Change directory to the matlab folder of your actual installed Gurobi version. The example below assumes 13.0.3.

TYPE IN: MATLAB Command Window

```
cd('C:/gurobi1303/win64/matlab')
gurobi_setup
which gurobi
```

2. Follow gurobi_setup instructions for saving the path (where appropriate). If MATLAB reports a different Gurobi path, resolve conflicting old versions before proceeding.

Gurobi MATLAB interface documentation

3. Run a direct, small Gurobi optimizer smoke test in MATLAB. It does not write study files:

TYPE IN: MATLAB Command Window

```
model.A = sparse(1);
model.obj = 1;
model.rhs = 1;
model.sense = '>';
model.lb = 0;
model.modelsense = 'min';
params.OutputFlag = 0;
sol = gurobi(model, params);
disp(sol.status);
assert(strcmp(sol.status,'OPTIMAL'));
```

> **PASS**
>
> which gurobi resolves to a Gurobi MATLAB interface, and the small model reaches OPTIMAL. A successful path check alone is insufficient: an optimization run additionally checks license access.

STAGE 08 / Connect the optimizer through MATPOWER

Goal: Verify that a DC-OPF case can be solved using the intended solver through MATLAB/MATPOWER.

Open: MATLAB Command Window

---

1. MATPOWER and Gurobi must both remain available on the path. Execute this independent small case test (not one of the publication experiments):

TYPE IN: MATLAB Command Window

```
mpopt = mpoption('verbose',0,'out.all',0, ...
    'opf.dc.solver','GUROBI');
mpc = rundcopf('case9', mpopt);
assert(mpc.success == 1);
disp(mpc.f);
```

2. If solver selection is rejected, check your MATPOWER/Gurobi installation versions and official solver integration documentation. Do not silently substitute a different optimizer when claiming identical numerical replication.

> **REPRODUCIBILITY NOTE**
>
> The study README records opf.dc.solver = DEFAULT and a separate current-environment GUROBI probe. The historical solver choice for every earlier run was not independently logged. The explicit GUROBI smoke test above is for the reader's current environment, not retrospective evidence.

STAGE 09 / SEGAN startup in MATLAB

Goal: Open the correct repository root and make SEGAN functions resolvable.

Open: MATLAB Command Window

---

1. Change MATLAB Current Folder to the extracted SEGAN repository root (where reproduce_all.m is visible). Do not start in src, data, or outputs.

TYPE IN: MATLAB Command Window

```
cd('C:/Research/SEGAN-State-Based-Screening')
addpath(fullfile(pwd,'config'),'-begin');
addpath(fullfile(pwd,'src'),'-begin');
which reproduce_all
which prepare_full_workspace
```

2. Check plan and stored validation without running Full Mode:

TYPE IN: MATLAB Command Window

```
plan = reproduce_all('plan');
disp(plan.SafeToStartFreshRun);
results = validate_results();
disp(results.Validation);
```

> **EXPECTED ROOT PROTECTION**
>
> The root contains twelve Quick checkpoints, so SafeToStartFreshRun may be false. This is intentional and not a failure. Full Mode requires an empty-output isolated workspace created later.

STAGE 10 / TRACK Q - reproduce article exports quickly

Goal: Regenerate the supported figures/tables from the twelve validated MAT checkpoints without rerunning OPF or clustering.

Open: MATLAB Command Window

---

1. Stay in the SEGAN root and confirm outputs/cache contains the twelve checkpoints. Do not start from the Full workspace.

2. Inspect the Quick plan before writing generated files:

TYPE IN: MATLAB Command Window

```
quickPlan = reproduce_all('quick_plan');
disp(quickPlan);
assert(quickPlan.SafeToStartQuickRun);
```

3. Only when the target folder is empty and the safety check passes, launch:

TYPE IN: MATLAB Command Window

```
quickReport = reproduce_all('quick');
disp(quickReport.Validation);
assert(quickReport.Validation.AllPassed);
```

4. Use File Explorer to open quick_reproduction/outputs/. Inspect the generated figure/table subfolders and exported CSV files. Expect 27 publication output files, counting image formats and plotting-data CSV.

SCREEN CHECKPOINT (illustrative; not an actual application capture)

WHAT YOU SHOULD SEE
quickPlan.SafeToStartQuickRun is true before run
quickReport.Validation.AllPassed is true on completion
quick_reproduction/outputs contains exported figures/tables and associated CSVs

PASS

Track Q has regenerated Figures 2-7 and Tables 4-9 from cached scientific checkpoints, without solving OPF or fitting K-means. Figure 1 was manually drawn and is not exported by this pipeline.

STAGE 11 / TRACK F - plan the full experiment

Goal: Prepare a safe, isolated workspace in which all computational stages can be rerun from the four raw inputs.

Open: MATLAB Command Window

1. Return to the original downloaded SEGAN root. Ensure MATPOWER and Gurobi from Stages 04-08 remain connected.

2. Execute the previously documented helper, first in read-only planning mode:

TYPE IN: MATLAB Command Window

```
addpath(fullfile(pwd,'config'),'-begin');
addpath(fullfile(pwd,'src'),'-begin');
workspacePlan = prepare_full_workspace('plan');
disp(workspacePlan);
assert(workspacePlan.SafeToCreate);
```

STOP CONDITION

If SafeToCreate is false, inspect the reported destination. Do NOT delete or overwrite an existing workspace to bypass the check. Use a fresh extract or move to a clean separate root according to USER_GUIDE.md.

STAGE 12 / TRACK F - generate a clean workspace

Goal: Copy four canonical inputs and reproducible source/configuration while omitting Quick cached outputs.

Open: MATLAB Command Window

---

1. When the previous plan passes:

TYPE IN: MATLAB Command Window

```
workspace = prepare_full_workspace();
disp(workspace.WorkspacePath);
cd(workspace.WorkspacePath);
addpath(fullfile(pwd,'config'),'-begin');
addpath(fullfile(pwd,'src'),'-begin');
clear functions;
rehash;
```

2. Confirm that the newly created Full workspace does not contain the twelve bundled Quick MAT checkpoints. The helper is designed to copy inputs and source/configuration, not precomputed Quick caches.

3. Inspect the Full Mode plan:

TYPE IN: MATLAB Command Window

```
plan = reproduce_all('plan');
disp(plan);
assert(plan.SafeToStartFreshRun);
```

> **PASS**
>
> A new empty-output Full workspace is active and SafeToStartFreshRun is true. The original root and Quick checkpoints remain unchanged.

STAGE 13 / TRACK F - compute all scientific stages

Goal: Run the SEGAN full pipeline from the four original data inputs.

Open: MATLAB Command Window

---

1. Within the isolated Full workspace created in Stage 12, enter:

TYPE IN: MATLAB Command Window

```
report = reproduce_all('full');
disp(report.Validation);
```

2. The supplied project documentation describes 23 computation/export stages, including DC-OPF cases, capacity-withholding scenarios, 57-bus feature libraries, clustering, economic/network metrics, robustness and publication exporters.

3. Expect newly generated .mat caches, figures, tables and CSV output files in the Full workspace's outputs tree. Do not copy the Full outputs on top of the original repository's prevalidated Quick files.

4. When Full completes, inspect the recorded validation; the author's reference clean-copy run previously passed 17/17 checks. A fresh external reproduction should report its own validation instead of assuming the author's outcome.

FULL MODE SUCCESS

All scheduled stages finish; no solver/license/runtime error remains; report.Validation reflects successful local checks. A different solver or software version must be documented if outputs differ.

STAGE 14 / Find the exact scientific outputs

Goal: Connect folders, figure/table files, numerical tables and scientific interpretations to the published article.

Open: File Explorer + MATLAB

Artifact	Where or how to find it
Figures 2-7	Quick: quick_reproduction/outputs/figures; Full: workspace outputs/figures
Tables 4-9	Quick: quick_reproduction/outputs/tables; Full: workspace outputs/tables
Numerical/plotting CSV	Generated under the corresponding output folders; filenames depend on export functions
Quick result inputs	Twelve shipped .mat files in outputs/cache
Full derived caches	Recomputed .mat files under the isolated Full workspace outputs/cache
Executable code	Root scripts, config/reproduction_config.m and src/+segan/*.m

Cross-check the exported file list and columns against the actual repo functions export_figures_and_tables.m, export_quick_results.m, and the 57-bus figure/table exporters. This manual intentionally does not invent exact exporter filenames not evidenced by the supplied README.

WHAT THE USER CAN RECREATE

Main scenario results for the 5-bus and 57-bus systems; Figures 2-7; Tables 4-9; the associated exported numerical CSVs. The manually authored Figure 1 and descriptive manuscript material require the article itself.

STAGE 15 / Understand the numerical metrics

Goal: Interpret what the exported values mean and avoid changing units or attribution assumptions.

Open: Publication + generated tables

Metric / model element	Interpretation
Reported-offer DC-OPF	Market clearing uses reported offers under imposed withholding scenarios
Baseline true-cost accounting	Ex-post cost/profit accounting is not identical to market-clearing offers
Capacity withholding screening	Exogenously imposed scenario analysis; not proof of intent or illegality
Nodal / economic outcomes	Network-constrained price, dispatch and accounting outputs from the validated pipeline
CCWDelta90Day, StrategicNE90Days	90-day sums, not daily averages
...MeanDaily	Daily averages, interpreted separately from 90-day totals
K-means cluster IDs	Arbitrary numeric labels; compare economic meaning and label-invariant metrics

The 57-bus workflow uses MATPOWER case57 with PGLib-SAD branch RATE_A substitutions and three 30-day, hourly windows based on 2022 Turkish demand shapes. The 5-bus focus includes G3; the 57-bus configurations are Single-G5 and Double-G5/G7. These are descriptions reported in the provided README, not independent reruns performed while preparing this PDF.

STAGE 16 / Document evidence of reproduction

Goal: Keep a transparent distinction between previously reported author checks and new third-party replication.

Open: MATLAB + researcher's notes

Evidence	How to report it
Installed software	MATLAB release/update, toolbox, MATPOWER, Gurobi exact versions
Solver selection	Settings, which gurobi / which rundcopf, solver smoke-test result
Input inventory	Four raw files and twelve Quick checkpoints; SHA-256 manifest
Quick result	Own quickReport.Validation and output folder/file inventory
Full result	Own report.Validation, stage completion and machine/solver details
Numerical comparison	Use project comparison/validation workflow; retain tolerances and audit logs
Reported prior evidence	Author: Full 17/17; Quick 27 output files; 27/27 CSV comparisons

The reported 27/27 numerical CSV comparisons are pairwise comparisons across original, Quick and clean Full records within the project's stated tolerances; not proof of byte-identical .mat, PDF or PNG files.

EVIDENCE BOUNDARY

A third party must actually run the code before declaring that the full experiment has been independently reproduced. This guide does not claim a new external execution or numerical audit.

STAGE 17 / Troubleshooting / recovery

Goal: Fix the specific step that failed without deleting established research results.

Open: Consult stage and component

Problem	Action
which rundcopf empty	In MATLAB, cd to MATPOWER root; run install_matpower; save path.
gurobi_setup not found	Open the installed Gurobi matlab folder for the correct installed version.
Gurobi license error	Check Named-User activation or WLS online access; do not expose credentials.
MATLAB/Gurobi incompatible	Check vendor support matrix; use a supported combination and report deviation.
SafeToStartQuickRun false	Target already occupied; inspect current outputs, use a clean extract; do not force deletion.
SafeToCreate false	Full workspace destination already exists; preserve it and use clean root.
Full fails while solving	Verify MATPOWER+Gurobi small model, license and selected solver, then log precise error.
Missing data	Check paths and filenames under data/raw, outputs/cache; compare manifest.
CSV values differ	Confirm seed, software versions, solver, preprocessing, reported tolerances and units.
Windows CMD vs MATLAB	CMD: dir, grbgetkey; MATLAB: addpath, gurobi_setup, reproduce_all.

Reference workflow: which tool and which command?

This page is designed to be kept beside MATLAB. For Windows users, the operating-system prompt and the MATLAB prompt must not be confused.

Phase	Tool / window	Action
Project files	Web browser	Code -> Download ZIP; extract the complete repository
Confirm inputs	Windows File Explorer or CMD	Verify four files in data/raw and twelve in outputs/cache
MATPOWER	MATLAB Command Window	cd to the MATPOWER root, then install_matpower
Gurobi key	Gurobi user portal + CMD	Retrieve personal grbgetkey command or configure WLS
Gurobi MATLAB API	MATLAB Command Window	cd to installed matlab folder, then gurobi_setup
SEGAN Quick	MATLAB Command Window	reproduce_all('quick_plan') -> reproduce_all('quick')
SEGAN Full	MATLAB Command Window	prepare_full_workspace('plan') -> prepare_full_workspace() -> reproduce_all('full')

DO NOT CONFUSE QUICK WITH FULL

Quick reproduces saved publication outputs from existing checkpoints and does not recompute the model. Full starts from the original raw inputs and executes the computational pipeline, including DC-OPF and clustering. These are different levels of reproducibility.

Run log template for an independent reproducer

[ ] OS + version: _____

[ ] MATLAB version/update + toolbox: _____

[ ] MATPOWER version + path: _____

[ ] Gurobi version, API path + license type (no secrets): _____

[ ] Quick run date, status + output file count: _____

[ ] Full run date, status + validation result: _____

[ ] Observed deviations from published settings: _____

The author previously reported 27/27 generated publication files in Quick Mode, 17/17 recorded-result validation checks in Full Mode, and 27/27 numerical CSV cross-run comparisons. Reproducers should distinguish these author checks from their own results.

End-to-end final checklist

[ ] Repository extracted with complete folder structure

[ ] Four canonical raw data inputs present

[ ] Twelve author-generated Quick Mode MAT files present

[ ] MATLAB + Statistics and Machine Learning Toolbox operational

[ ] MATPOWER 8.1 linked and DC-OPF smoke test successful

[ ] Gurobi installed and academic license activated

[ ] MATLAB Gurobi API test reaches OPTIMAL

[ ] Quick Mode validation and publication exports reviewed

[ ] Full workspace isolated and plan safe

[ ] Full Mode pipeline and local validator completed

[ ] Figures 2-7 and Tables 4-9 located

[ ] Research metrics, units and limitations interpreted

[ ] Research software/data citation, source attribution and sharing terms observed

> **PUBLICATION AND SHARING**
>
> The MATLAB implementation is identified by the project as author-licensed under BSD-3-Clause, whereas the third-party-linked inputs have separate rights considerations. The GitHub repository was private at the time the supplied materials were prepared; public downloads require the owner to publish it. The user's aim is a self-contained package, but no guide can independently grant upstream redistribution rights.

Authoritative installation and licensing references

MATPOWER 8.1 installation

MATPOWER installer reference

Gurobi MATLAB setup

Gurobi academic licensing

Gurobi MATLAB compatibility

Named-User grbgetkey

Gurobi academic WLS restrictions

MathWorks installer

Published SEGAN paper

Methodology note: SEGAN-specific command lines and reported numerical results are grounded in the provided README.md and earlier conversation evidence. Installation/license guidance uses the cited official provider documentation as of 04 Nov 2026. Any schematic check panels are explicitly illustrative rather than photographed application screenshots.